Questions

Q1.

A sequence a_1, a_2, a_3, ... is defined by

$$a_{n+1} = 4 - a_n$$

$$a_1 = 3$$

Find the value of

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(a) (i) a_2

(ii) a_{107}

(2)

(b) $\sum_{n=1}^{200}(2a_n - 1)$

(2)

(Total for question = 4 marks)

(Q01 WMA12/01, June 2019)

Q2.

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A sequence α_1, α_2, α_3, ... is defined by

$$a_{n+1} = 2(a_n + 3)^2 - 7$$

$$a_1 = p - 3$$

where p is a constant.

(a) Find an expression for α_2 in terms of p, giving your answer in simplest form.

(1)

Given that an

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$$\sum_{n=1}^{3} a_n = p + 15$$

(b) find the possible values of α_2

(6)

(Total for question = 7 marks)

(**Q08 WMA12/01, Jan 2021**)

Q3.

A sequence u_1 , u_2 , u_3 , ... is defined by

$$u_1 = 3$$

$$u_{n+1} = 2 - \frac{4}{u_n}$$

(a) Find the value of u_2 , the value of u_3 and the value of u_4

(3)

(b) Find the value of

$$\sum_{r=1}^{100} u_r$$

(2)

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(Total for question = 5 marks)

(**Q02 WMA12/01, Oct 2023**)

Q4.

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(i) An arithmetic series has first term α and common difference d.

Prove that the sum to n terms of this series is

$$\frac{n}{2}\{2a+(n-1)d\}$$

(3)

(ii) A sequence u_1, u_2, u_3,... is given by

$$u_n = 5n + 3(-1)^n$$

Find the value of

(a) u_5

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(1)

(b) $\sum_{n=1}^{59} u_n$

(3)

(Total for question = 7 marks)

(Q08 WMA12/01, Jan 2020)

Q5.

A sequence u_1, u_2, u_3, ... is defined by

$$u_n+1 = b - au_n$$

$$u_1 = 3$$

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where a and b are constants.

(a) Find, in terms of a and b,

(i) u_2

(ii) u_3

(2)

Given

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- $\sum_{n=1}^{3} u_n = 153$
- $b = a + 9$

(b) show that

$$a^2 - 5a - 66 = 0$$

(3)

(c) Hence find the larger possible value of u_2

(3)

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(Total for question = 8 marks)

(Q11 WMA12/01, June 2023)

Q6.

A sequence a_1, a_2, a_3, ... is defined by

$$a_n = \cos^2\left(\frac{n\pi}{3}\right)$$

Find the exact values of

(a) (i) a_1

(ii) a_2

(iii) a_3

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(3)

(b) Hence find the exact value of

$$\sum_{n=1}^{50} \left\{ n + \cos^2\left(\frac{n\pi}{3}\right) \right\}$$

You must make your method clear.

(4)

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(Total for question = 7 marks)

(Q03 WMA12/01, Oct 2022)

Q7.

(i) Find the value of

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$$\sum_{r=1}^{\infty} 6 \times (0.25)^r$$

(3)

(ii) A sequence u_1, u_2, u_3,... is defined by

$$u_1 = 3$$

$$u_{n+1} = \frac{u_n - 3}{u_n - 2} \qquad n \in \mathbb{N}$$

(a) Show that this sequence is periodic.

(2)

(b) State the order of this sequence.

(1)

(c) Hence find

$$\sum_{n=1}^{70} u_n$$

(2)

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(Total for question = 8 marks)

(Q05 WMA12/01, Jan 2024)

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